Vertical GaN Power Devices

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Abstract

GaN devices, in the form of the lateral AlGaN/GaN HEMTs, have shown great performance in medium power applications. However, to address the higher voltage applications (>1200V), devices with vertical geometry are necessary. In this paper, the progress made in vertical GaN technology has been reviewed. Key barriers, such as Mg ion implantation and regrowth, have been discussed.

(Keywords: GaN, Power Electronics and Vertical Power Device)

Introduction

Silicon power devices have served the power electronics industry for over 50 years with device performances are reaching the material limits. With the increasing requirements on more efficient power electronics, exploring advanced materials is necessary. Wide bandgap (WBG) semiconductors, including gallium nitride (GaN) and silicon carbide (SiC), are the material candidates for next generation power switches. GaN has a critical electric field of 3 MV/cm and electron's mobility over 1200 cm^2/Vs, both of which enable GaN a promising candidate for power switch applications. Due to the nature of high electron density and mobility in AlGaN/GaN heterostructures, AlGaN/GaN based high electron mobility transistors (HEMTs) have been developed and shown excellent performance in medium power applications. However, limited by the lateral geometry, the HEMT needs complicated field plate structure to reduce the local electric field peaks, which make it undesired for high voltage applications (>1200V). For high power applications, a vertical device architecture is preferred since it offers high breakdown electric field close to the theoretical value and the higher current density.

A Survey of Vertical GaN Devices

A. Current Aperture Vertical Electron Transistor

The current aperture vertical electron transistor (CAVET) is a vertical device that is a combination of the AlGaN/GaN HEMT and the DMOS structures, as shown in Fig. 1. The device features with an extreme high channel electron mobility due to the polarization-induced two-dimensional electron gas (2DEG) in the AlGaN/GaN heterojunction interface. Moreover, the vertical p-n junction formed by the p-GaN current blocking layer and the n-GaN drift region enables a high voltage blocking capability. The very early GaN CAVET was developed for RF application [1]. Chowdhury et al demonstrated its potential for high power applications [2]. Several research groups have reported CAVET with blocking voltages of 600V to 1700V and the specific on-state resistance less than $3m\Omega\cdot cm^2$ [3-6]. Although the polarization-induced 2DEG in AlGaN/GaN interface makes the AlGaN/GaN-based channel normally on, several techniques have been proposed to realize the normally-off operation: the p-GaN gate CAVET structure was adopted by Ref. 3 and 4, and the cascoded CAVET structure was proposed by Ref. 7.

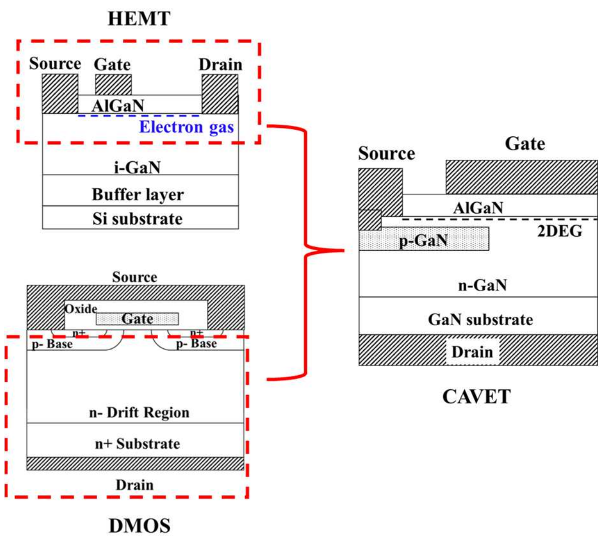

Fig. 1 The schematic of the CAVET structure, which is a combination of HEMT and DMOS.

B. MOSFET

GaN-based MOSFET was developed to realize a robust normally-off operation. Limited by the selective-area doping technique, the trench MOSFET,

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as shown in Fig. 2 (a), was firstly developed [8]. Although excellent volage blocking capability was demonstrated in the GaN trench MOSFETs [9-11], the on-state resistance is limited by the poor channel electron mobility. Gupta et al developed an in-situ oxide, GaN interlayer-based trench MOSFET (OG-FET), which enhanced the channel electron mobility without sacrificing the normally-off operation [12]. Fig. 2 (b) illustrates the OG-FET structure. A channel electron mobility of 185 cm^2/Vs was demonstrated in the OG-FET [13]. In the past few years, a progress has been made in p-type GaN formed by Mg ion implantation, which makes VDMOS structure available, as shown in Fig. 2 (c). A recent study has demonstrated 1.2 kV/1.4 mΩ·cm^2 performance in the GaN VDMOS with the base region formed by Mg ion implantation [14]. Another MOSFET structure is fin MOSFET, as shown in Fig. 2 (d), which has no p-type GaN layers. Li et al. simulated and demonstrated the concept in 2016 [15], and Zhang et al. has demonstrated high performance 1.2kV-class fin MOSFET in 2017 [16].

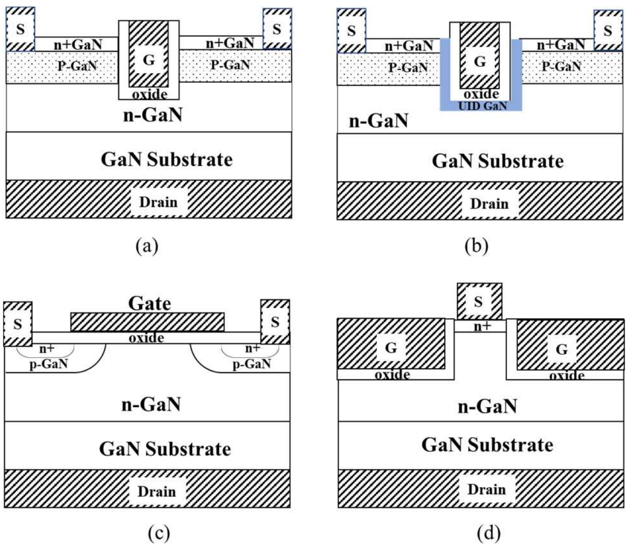

Fig. 2 (a) Trench MOSFET; (b) Trench MOSFET with UID GaN inserting layer (OG-FET); (c) Planar gate VDMOS; and (D) Fin MOSFET.

C. Junction Field-Effect Transistor

Junction field-effect transistors (JFETs) are devices with the n-type channel enclosed by the gate formed by p-type GaN. The conductivity of the channel can be modulated by changing the depletion width of the gate-to-channel p-n junction, the structures of JFETs are illustrated in Fig. 3. Both lateral channel and vertical channel GaN JFETs were proposed and simulated by Ji et al. [17], and then 1.2kV JFETs were demonstrated by Avogy Inc., [18]. Till now, 1.2-kV class vertical channel JFET has been demonstrated with robust avalanche capability [19], which is a big progress to move forward to the commercialization.

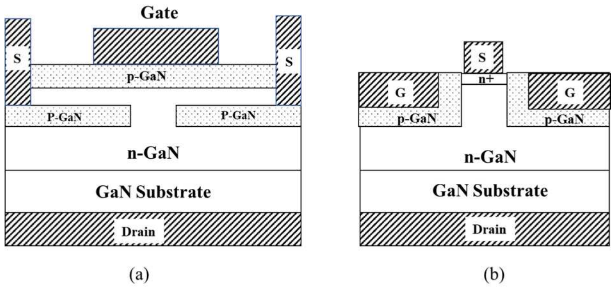

Fig. 3 (a) Lateral channel JFET; (b) Vertical channel JFET.

D. Static Induction Transistor

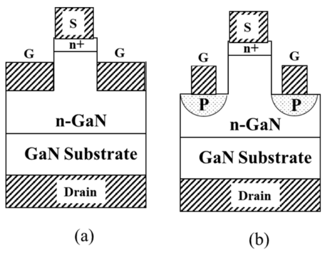

Fig. 4 (a) Schottky-gate static induction transistor; (b) p-GaN gate static induction transistor.

The static induction transistor (SIT) has a unique conductivity modulation mechanism different from the field-effect transistors. It utilizes a carrier injection mechanism through the space charge region induced by the gate bias, so the device has a unique triode-like I_D-V_{DS} characteristics. Li et al. demonstrated the GaN-based Schottky-gate SIT (as shown in Fig. 4 (a)) adopting a self-aligned process [20], and Chun et al. demonstrated the high voltage blocking capability of the device [21]. Another variation of the SIT structure is the p-GaN gate SIT, which is shown in Fig. 4 (b). Compared to the Schottky-gate SIT, the p-GaN gate one is more favorable for high power applications.

E. Diodes

Power diodes are also important devices in power electronics. Depends on the voltage ratings, there are typically three power diode structures: the Schottky barrier diode (SBD) (Fig. 5 (a)) for medium and low voltage applications, the junction barrier Schottky (JBS) diode (Fig. 5 (b)) for high voltage applications, and the P-I-N diode (Fig. 5 (c)) for ultra-high voltage applications. In the past few years, high voltage GaN diodes with breakdown voltage up to 5 kV were demonstrated by multiple research groups [22-33]. Moreover, robust avalanche breakdown in vertical GaN PIN diodes were also demonstrated [22, 23, 28, 29, 31]. Another important achievement is the measurement of impact ionization coefficients of GaN using PN diodes, which enables the accurate projection of the critical electric field of GaN layers with different doping concentration [34-37].

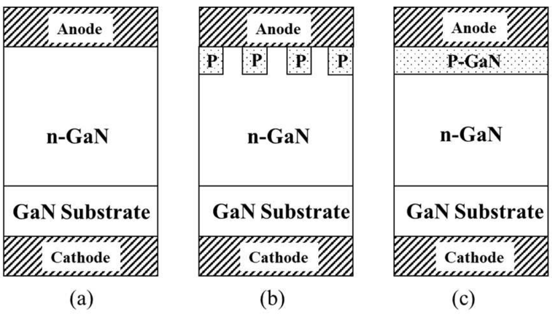

Fig. 5 (a) Schottky barrier diode (SBD); (b) Junction barrier Schottky diode (JBS Diode); (c) P-i-N diode.

Key Challenges

One of the barriers for commercialization of vertical GaN devices is the high cost of free-standing GaN substrate. Currently, the 2-inch and 4-inch GaN substrates are commercialized, but the 4-inch substrate is not cost-efficiently for large scale power device manufacturing.

Another barrier is the unmatured GaN fabrication techniques. The most obvious one is p-GaN formation by ion implantation. The post-implantation annealing process require a high temperature over 1400 °C, at which temperature the GaN surface suffers from decomposition. Cap layers, such as AlN, have been developed to protect the GaN surface, and a substantial amount of Mg ions can be activated during the post implantation annealing in ultra-high pressure and high temperature conditions [38]. Another challenge is the regrowth process. CAVETs and JFETs need the regrowth process, removal of the interface impurities needs more study [39].

Conclusion

In summary, this article presents various vertical GaN device structures for power applications, including CAVET, MOSFET, JFET, SIT and the diodes. Key challenges of vertical GaN technology were discussed.

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